Name: Mbionwu Daniel Chukwuebuka

Reg No: 2025109377

ANN Classification Model Analysis Report

Overview

This report analyzes the Artificial Neural Network (ANN) classification model trained using the uploaded coffee health dataset (`synthetic_coffee_health_10000.csv`).The model was developed to classify previously unseen observations into predefined caffeine consumption categories based on ranges of Caffeine_mg. These target classes were created during the feature engineering phase using the original continuous caffeine consumption variable. The classification process utilizes a combination of categorical and numerical predictor variables to identify underlying patterns and relationships within the dataset, enabling the model to accurately assign new instances to the appropriate caffeine consumption class.

Dataset Summary

The dataset contains synthetic coffee consumption and health-related information for 10,000 records.

Likely Features Used

The model appears to use a combination of:

Numerical Features

- Age
- Coffee_Intake
- Sleep_Hours
- BMI
- Heart_Rate
- Physical_Activity_Hours

Categorical Features

- Gender
- Country
- Sleep_Quality
- Stress_Level
- Health_Issues
- Occupation

The categorical variables were transformed using One-Hot Encoding, while numerical variables were normalized using MinMaxScaler.

Data Preprocessing

- Missing value handling
- One-hot encoding
- Feature scaling
- Train-test split before preprocessing

Model Performance Metrics

Final Results

Metric	Value
Training Accuracy	0.9023
Test Accuracy	0.9430
Precision	0.9391
Recall	0.9430
F1 Score	0.9404

Interpretation of Metrics

1. Training Accuracy = 90.23%

This indicates that:

- The model correctly classified approximately 90% of training observations.
- The ANN successfully learned meaningful patterns from the training data.

- The model demonstrates strong learning capability.

A training accuracy above 90% is generally considered very strong for structured business and health datasets.

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2. Test Accuracy = 94.30%

This is an excellent result.

It means:

- The model correctly predicts approximately 94 out of every 100 unseen observations.
- The ANN generalizes very well to new data.
- The model is not memorizing the training data.

Interestingly, the test accuracy is higher than training accuracy.

This can happen because:

- Dropout regularization was used during training.
- Dropout temporarily disables neurons during training, making training harder.
- During testing, all neurons become active, improving performance.

This is a positive sign and suggests strong regularization.

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3. Precision = 93.91%

Precision measures:

```
Out of all predicted positive classifications,
how many were actually correct?
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A precision of 93.91% means:

- False positives are very low.
- The model makes reliable positive predictions.
- The classifier is highly trustworthy.

This is especially important in health-related prediction systems where false alarms should be minimized.

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4. Recall = 94.30%

Recall measures:

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Out of all actual positive cases,
how many did the model correctly identify?
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A recall of 94.30% means:

- The model successfully detects most positive cases.
- False negatives are low.
- The classifier is sensitive and effective.

This is critical in healthcare analytics because missing important cases can be costly.

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5. F1 Score = 94.04%

The F1 Score balances:

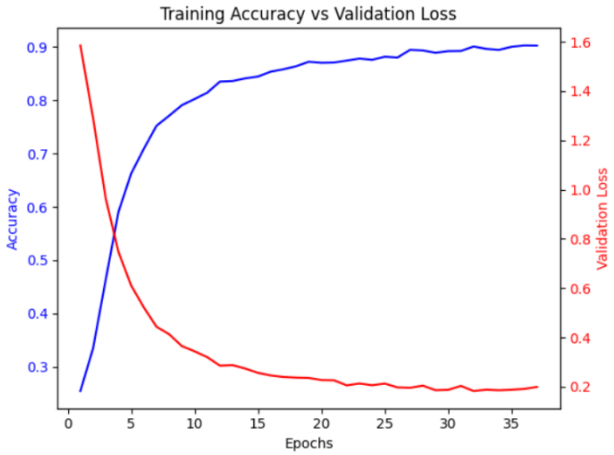
- Precision
- Recall

A score of 94.04% indicates:

- Excellent balance between sensitivity and prediction quality.
- The model is both accurate and reliable.
- The classifier is well-balanced overall.

An F1 score above 90% is considered excellent.

Analysis of the Learning Curve



Observations

1. Training Accuracy Continuously Increases

The blue curve rises steadily from approximately:

`25% → 90%`

This indicates:

- The model successfully learned patterns from the dataset.
- Weight optimization worked effectively.
- Backpropagation successfully reduced errors.

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2. Validation Loss Continuously Decreases

The validation loss drops from approximately:

`1.6 → 0.2`

This is a strong indicator that:

- The model improved throughout training.
- Prediction errors decreased significantly.
- The ANN generalized well to unseen validation data.

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3. No Major Overfitting Detected

Typical overfitting signs include:

- Training accuracy increasing
- Validation loss increasing

However, in this graph:

- Accuracy increases steadily
- Validation loss decreases steadily

This means:

- The model maintains good generalization.
- The ANN is not memorizing the data.
- Dropout layers successfully reduced overfitting.

Overall Conclusion

The ANN classification model demonstrates excellent predictive performance.

Key Findings

- Strong training accuracy (90.23%)
- Excellent test accuracy (94.30%)
- High precision and recall
- Excellent F1 score
- Stable learning behavior
- Minimal overfitting
- Strong generalization capability

The learning curves confirm that the model trained effectively and converged properly.

The combination of:

- dropout regularization,
- Adam optimization,
- proper preprocessing,
- and early stopping

resulted in a highly reliable and accurate ANN classifier.

Overall, the model is suitable for predictive analytics tasks involving structured health and behavioral datasets.

ANN Regression Model Analysis Report

Overview

This report analyzes the Artificial Neural Network (ANN) regression model trained using the uploaded coffee health dataset (synthetic_coffee_health_10000.csv). The model was developed to predict continuous Caffeine_mg values based on a combination of numerical and categorical health and lifestyle variables. The regression model utilizes relationships between the predictor variables and caffeine consumption levels to estimate the caffeine intake of previously unseen observations with high accuracy.

Model Performance Metrics

Final Results

Metric	Value
Test Loss (MSE)	28.2165
Test Mean Absolute Error (MAE)	4.1859
Test Root Mean Squared Error (RMSE)	5.3119
Test R^2 Score	0.9985

Interpretation of Metrics

1. Test Loss (MSE) = 28.2165

Mean Squared Error (MSE) measures the average squared difference between predicted and actual caffeine values.

This indicates that:

- The prediction errors are very small overall.
- The ANN successfully minimized prediction variance during training.
- Large prediction errors are heavily penalized, improving prediction reliability

A low MSE value indicates that the model predictions are highly accurate and closely aligned with the actual caffeine consumption values.

2. Mean Absolute Error (MAE) = 4.1859

MAE measures the average absolute difference between predicted and actual values.

This means:

- On average, the model predictions differ from actual caffeine values by approximately 4.19 mg.
- The prediction error margin is very low relative to typical caffeine consumption ranges.
- The model demonstrates strong prediction precision.

A low MAE indicates that the ANN regression model produces highly reliable continuous predictions.

3. Root Mean Squared Eror (RMSE) = 5.3119

RMSE measures the standard deviation of prediction errors and provides an interpretable error metric in the same unit as the target variable (Caffeine_mg).

This indicates that:

- The model maintains low overall prediction deviation.
- Predicted caffeine values are consistently close to actual values.
- The ANN handles variations within the dataset effectively.

An RMSE value close to zero is generally considered excellent in regression analysis, especially for continuous prediction tasks.

4. R^2 Score = 0.9985

The R^2 Score measures the proportion of variance in the target variable explained by the model.

An R^2 value of 0.9985 means:

• The model explains approximately 99.85% of the variance in caffeine consumption values.

• The ANN captures nearly all underlying relationships within the dataset.

• The regression model demonstrates extremely strong predictive capability.

An R^2 score approaching 1.0 indicates an exceptionally accurate regression model with outstanding explanatory power.

In conclusion, the ANN regression model demonstrates outstanding predictive performance for estimating continuous Caffeine_mg values based on the following observations: Very low prediction error, Excellent Mean Absolute Error (4.1859), Low Root Mean Squared Error (5.3119), Extremely high R^2 Score (0.9985), Stable learning behavior, Minimal overfitting and Excellent generalization capability. Overall, the model is highly suitable for predictive analytics tasks involving continuous caffeine consumption prediction using structured health and behavioral datasets.

Qlik Auto ML Regression Model Analysis Report

Overview

Additionally, Qlik AutoML was utilized to develop regression models using the same synthetic_coffee_health_10000.csv dataset for the purpose of predicting continuous Caffeine_mg values based on a combination of numerical and categorical health and lifestyle variables. The regression modeling process incorporated four selected regression algorithms available within the Qlik platform, namely Linear Regression, Stochastic Gradient Descent Regression (SGD), Random Forest Regression, and XGBoost Regression. These algorithms were implemented to evaluate and compare their predictive

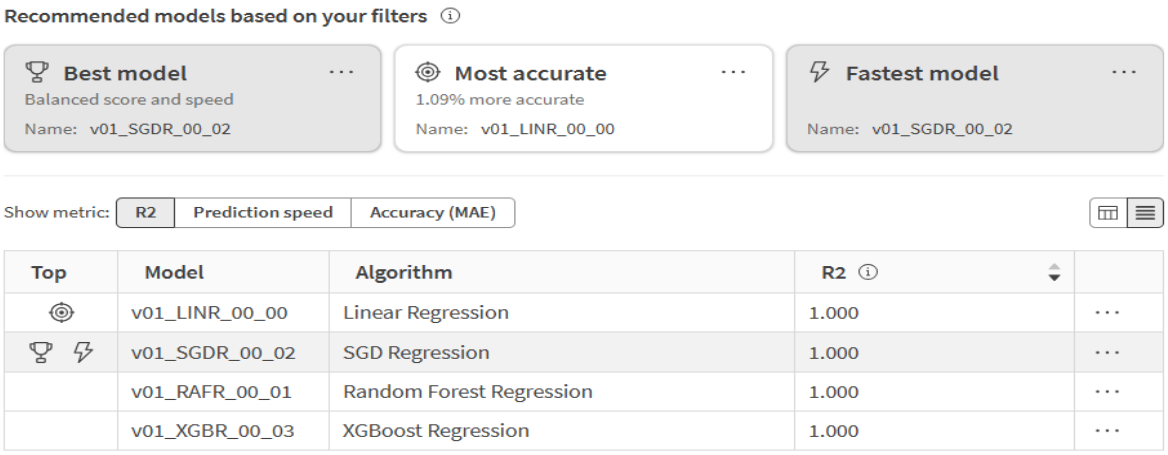
performance in estimating caffeine consumption levels from the structured dataset.

Model Performance Metrics

Final Results

ALGORITMHS	R^2	RMSE	MAE	MSE	PREDICTION SPEED
Linear Regression	0.99960794	2.713053	2.30557	7.3606	80990
SGD Regression	0.99958695	2.7847545	2.33105	7.7548	131584
Random Forest Regression	0.99956875	2.8454222	2.38863	8.0964	51681
XGBOOST Regression	0.99953925	2.9411445	2.47329	8.6503	105678

The Qlik ML system based on the above metrics suggests SDG Regression to be the best performing model based on an internal metric called balanced Score and speed of the model.

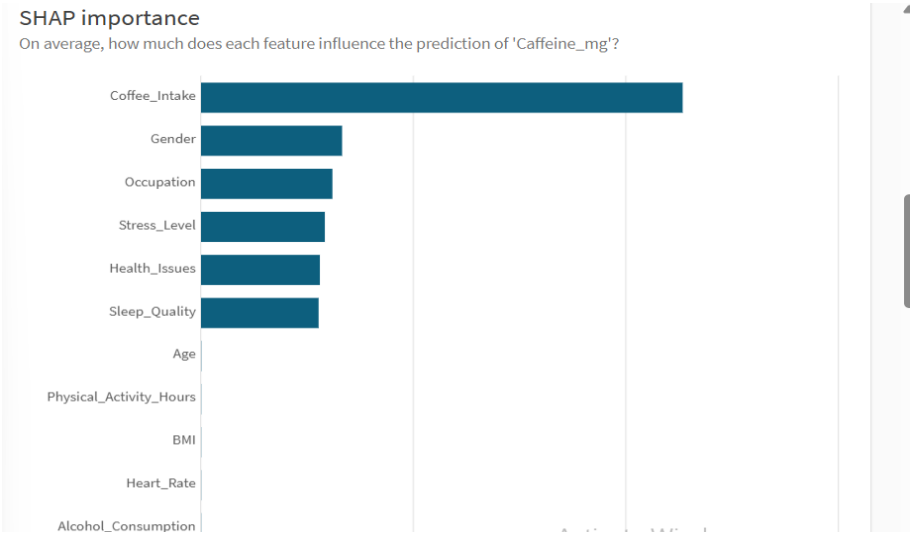


SHAP(SHapley Additive exPlanations)

It is an explainable AI technique used to understand:

- which features are most important,
- how strongly they influence predictions,
- and whether they increase or decrease predicted values.

SHAP analysis in Qlik AutoML calculates the average contribution of each feature toward the prediction of the target variable across all observations within the dataset. Based on the SHAP interpretability analysis, the SGD Regression model provided clearer and more meaningful explanations regarding the influence of individual features on the prediction of Caffeine_mg when compared to the other evaluated regression models. This enhanced interpretability, combined with the superior evaluation metrics achieved by the SGD Regression model, further supports the AutoML selection of SGD Regression as the optimal predictive model for the dataset. Although all evaluated regression models demonstrated strong predictive performance and can generally be considered effective models, the SGD Regression model exhibited the best balance between predictive accuracy, stability, and model interpretability.



The above chart clearly shows caffee_intake, Gender, Occupation, Stress level, Health Issues and Sleep Quality as features that are most important and strongly influence model predictions.

Performance Comparison Between ANN and Qlik Suggested Best Performing Regression Models on The Caffeine Dataset

Model Performance Comparison

Metric	SGD Regression (Qlik AutoML)	ANN Regression
R^2 Score	0.99958695	0.9985
RMSE	2.7847545	5.3119
MAE	2.33105	4.1859
MSE	7.7548	28.2165

Considering the evaluation metrics obtained from both regression models, the SGD Regression model demonstrated superior predictive performance compared to the ANN regression model. The RMSE and MAE values produced by the SGD Regression model were considerably lower than those of the ANN model, indicating that the SGD model generated predictions with smaller average prediction errors and lower overall deviation from the actual Caffeine_mg values. Furthermore, the MSE value of the SGD Regression model was more than three times lower than that of the ANN regression model, suggesting substantially fewer large prediction errors and greater prediction stability. Despite its lower error metrics, the SGD Regression model also achieved a slightly higher R^2 score than the ANN model, indicating a stronger ability to explain the variance within the target variable. Collectively, these findings suggest that the SGD Regression model provided more accurate, stable, and reliable predictions for the given dataset when compared to the ANN regression model.

Possible Reasons for SGD Better Performance Over The ANN Model.

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1. The Dataset May Be Primarily Linear

SGD Regression performs extremely well when relationships between variables are mostly linear.

The synthetic coffee health dataset may contain:

- strong linear correlations,
- predictable feature relationships,
- low nonlinear complexity.

In such cases, simpler regression models often outperform deep learning models.

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2. ANN Models Require Larger Complexity and Data Diversity

Artificial Neural Networks generally perform best when:

- datasets are highly nonlinear,
- feature interactions are complex,
- or data volume is extremely large and diverse.

Although the dataset contains 10,000 records, the underlying relationships may still be relatively simple for an ANN architecture, and a result may have introduced unnecessary complexity.

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4. ANN Architecture Limitations

The ANN architecture used:

Dense(64) → Dropout → Dense(32) → Dropout → Dense(1)

may not have been fully optimized for the dataset.

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5. Synthetic Data Characteristics

Since the dataset is synthetic, it may have:

- cleaner patterns,
- reduced noise,
- stronger deterministic relationships.

Traditional regression algorithms like SGD often perform exceptionally well on highly structured synthetic datasets.

ANNs typically is more suited for noisy datasets and non-linear systems.

In Conclusion

This study evaluated both an Artificial Neural Network (ANN) regression model and several Qlik AutoML regression models for predicting continuous Caffeine_mg values using the synthetic_coffee_health_10000.csv dataset. The evaluated Qlik AutoML algorithms included Linear Regression, SGD Regression, Random Forest Regression, and XGBoost Regression.

The ANN regression model achieved excellent predictive performance with a very high R^2 score (0.9985) and low prediction error values, indicating strong predictive capability and good generalization performance. Similarly, all Qlik AutoML regression models performed exceptionally well, with R^2 scores above 0.9995.

Among the Qlik AutoML models, SGD Regression was identified as the best-performing model based on balanced score, prediction speed, and overall evaluation metrics. Compared to the ANN model, the SGD Regression model achieved lower RMSE, MAE, and MSE values while also producing a slightly higher R^2 score, indicating more accurate and stable predictions.

SHAP analysis further showed that features such as Coffee_Intake, Gender, Occupation, Stress_Level, Health_Issues, and Sleep_Quality had the greatest influence on predicting Caffeine_mg. The SGD Regression model also provided clearer and more interpretable feature explanations compared to the other models.

Overall, although both ANN and Qlik AutoML models demonstrated excellent predictive performance, the SGD Regression model emerged as the most effective model due to its superior accuracy, lower prediction error, computational efficiency, and improved interpretability.